Online NLP and text mining tools and examples

Tokenization

- deciding what constitutes a term
- splitting a text into tokens
- might seem simple at the first glance (e.g., for English) – words are separated by spaces (unlike, e.g., in Chinese)
- a good tokenizer must handle punctuation (e.g., don't, Jane's, and/or), hyphenation (e.g., state-of-the-art), or recognize multi-word terms (e.g., *Barack Obama* and *ice hockey*)

Tokenization

- http://text-processing.com/demo/tokenize/

- try to tokenize your own sentences and the following ones:
  - On 2018-02-14, I had a long-lasting phone call from 123-555-9999. (hyphenated expressions)
  - The dr. lives in a tall building in New York, U.S.A. (abbreviations)
  - Don't look at me. (apostrophe)
  - Mail me at qqq@gmail.com. (entity)
  - The price is $1,234.56. (number)

Stemming

- the process of reducing inflected words to their stems
- in English, affixes are simpler and more regular than in many other languages -> stemming algorithms based on heuristics work relatively well
- Porter stemming: ED -> ‘’ (plastered -> plaster), ATIONAL -> ATE (relational -> relate), ATOR -> ATE (operator -> operate), EMENT -> ‘’ (replacement -> replac) etc.
- Snowball – a language for creating stemmers (https://snowballstem.org/, http://snowball.tartarus.org/)

Stemming

- http://text-processing.com/demo/stem/

- prepare a few sentences and see how the words are stemmed

Part-of-speech tagging

- labeling words according to their parts of speech, i.e., determining whether a word is a noun, verb, adjective, adverb, pronoun, preposition, conjunction, interjection, numeral, or article

- https://corenlp.run/

- http://www.ling.upenn.edu/courses/Fall_2003/ling001/penn_treebank_pos.html
  - alphabetical list of part-of-speech tags

Parsing

- analyzing the grammatical structure of sentences and marking the words in the sentences according to their grammatical roles
- relations of a word to others and the functions of the words (subject, object, etc.) become obvious

- https://corenlp.run/
- http://www.link.cs.cmu.edu/link/submit-sentence-4.html

- try to submit a few sentences and see the results:
  - part-of-speech tags
  - links between pairs of words
  - parse tree

Parsing

The parsed sentence: *The parser expects just one sentence*.

connects a transitive verb to an object

connects a singular determiner to a noun

```
    +------------------------Xp------------------------+
    +------Wd-----+        +----------Os---------+     |
    |      +--Ds--+---Ss---+        +-EN-+---Ds--+     |
    |      |      |        |        |    |       |     |
LEFT-WALL the parser.n expects.v just.e one sentence.n .
```

connects an adverb to an expression of quantity

```
Constituent tree:
```

noun verb adverb

```
(S (NP The parser)
   (VP expects
       (NP just one sentence))
   .)
```

sentence verb phrase noun phrase

Parsing – syntax categories

Symbol	**Meaning**	**Example**
S	sentence	the man walked
NP	noun phrase	a dog, almost every sentence
VP	verb phrase	saw a park, finished the work
PP	prepositional phrase	with a telescope
Det	determiner	the
N	noun	dog
V	verb	walked
P	preposition	in

Source: http://www.nltk.org/book/

Sentiment analysis

- assigning sentiment to documents or aspects
- http://text-processing.com/demo/sentiment/
- SentiStrength: https://mi-linux.wlv.ac.uk/~cm1993/sentistrength/
    - prepare a few sentences – neutral, positive, and negative, with different levels of emotion (e.g., strongly and slightly negative).
    - ty to quantify their sentiment using your own judgment on a scale from -5 to +5.
    - try to detect sentiment for the sentence "There were mice in my room." in the context of music and hotels.

Sentiment analysis

- Try to assign a sentiment level to the following sentences by yourself. Then detect dual/binary/scale sentiment for the sentences using SentiStrength. Compare the results and explain the differences.
  - I really love you but dislike your cold sister.
  - Most automated sentiment analysis tools are shit.
  - Sentiment analysis has never been good.
  - I won't say that the movie is astounding and I wouldn't claim that the movie is too banal either.
  - This movie was actually neither that funny, nor super witty.

Named entity recognition

- finding named entities and categorizing them (people, places, organizations, …)

- Stanford Named Entity Tagger
- https://corenlp.run/

- try to find a few news (e.g., on http://www.telegraph.co.uk/news/) or prepare your own text and let the tagger identify the entities

Co-reference resolution

- finding pieces of text that refer to the same entity
- https://corenlp.run/
- explanation of the task and sample texts here https://web.stanford.edu/class/archive/cs/cs224n/cs224n.1162/handouts/cs224n-lecture10-coreference.pdf or here https://www.linkedin.com/pulse/coreference-resolution-natural-language-processing-nlp-r-poykc/

Open information extraction

- extraction of verbs and their arguments (typically)
  - working with POS tags (e.g., ReVerb – http://reverb.cs.washington.edu/)
  - working also with dependency parsing (e.g., ClausIE)
- ClausIE
  - https://www.mpi-inf.mpg.de/departments/databases-and-information-systems/software/clausie/
  - S: Subject, V: Verb, C: Complement, O: Direct object, O_i: Indirect object, A: Adverbial, V_i: Intransitive verb (*has an object*), V_c: Copular verb (expresses that the subject and its complement denote the same thing or that the subject has the property denoted by its complement), V_{mt}: Monotransitive verb (*has a single direct object*), V_{dt}: Ditransitive verb (*has a direct object and an indirect object*), V_{ct}: Complex-transitive verb (*requires a direct object and another object or an object complement*)

Open information extraction

- see what phrases are identified by ClausIE (or https://corenlp.run/) in the following sentences
  - Mendel University is located in Brno.
  - The remainder of this paper is structured as follows.
  - A clause is part of a sentence that expresses some coherent piece of information.
  - We propose ClausIE, a novel, clause-based approach to open information extraction, which extracts relations and their arguments from natural language text.
  - He fathered two children, Edna and Donald, and lived in Aberdeen until his death from tuberculosis in 1942.

WordNet

- a lexical database of English
- nouns, verbs, adjectives, and adverbs are grouped into sets of cognitive synonyms (synsets), each expressing a distinct concept
- synsets are interlinked by means of conceptual-semantic and lexical relations
    - super-subordinate relation (also called hyperonymy or hyponymy)
    - meronymy = the part-whole relation
- https://wordnet.princeton.edu/

Word similarity

- http://vectors.nlpl.eu/explore/embeddings/en/

- computing the word similarity between two words by word2vec

- try to find the most similar words to a given word, e.g., *university*
- compute word analogy, e.g.: *man*, *king*, *woman* or *Germany, Berlin, Czech*

Question answering

- Stanford Question Answering Dataset (SQuAD)
  - SQuAD 1.1 - 100,000+ question-answer pairs on 500+ articles
  - SQuAD2.0 - SQuAD1.1 with over 50,000 unanswerable questions
  - https://rajpurkar.github.io/SQuAD-explorer/

Summarization

- https://quillbot.com/summarize

NLP studio

- developed at MENDELU
- selected preprocessing and processing tasks using visual programming
- https://mta.pef.mendelu.cz/lab/

VecText

- application for transforming texts to traditional structured representation
- graphical and command line interface
- https://sourceforge.net/projects/vectext/
- paper: http://www.iaeng.org/IJCS/issues_v46/issue_2/IJCS_46_2_05.pdf